Problem 10 - A falling chain

10. **A falling chain.** [10 points]

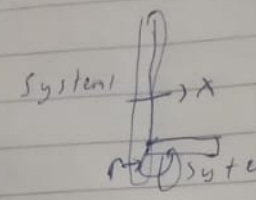
You hold a chain of mass M, length L and uniform mass density λ from one end above a fixed table (right figure above). The chain falls under constant gravitational acceleration. Ignore any swinging or bouncing in the chain and assume that it moves purely in a vertical direction until it comes to rest hanging straight down below the table.

(10 points) What is the support force from the table, F_S, as a function of time after the chain is released?

Solution (handwritten) follows on the next pages.

Problem 10. (Hanging Chain)

Assumptions :- 1) Top point falls with $-g$ acceleration
2) Extreme Turning point



$L = x + 2r$

$r = \frac{L-x}{2} \quad \frac{dr}{dt} = \frac{-1}{2}\frac{dx}{dt} = +\frac{gt}{2}$

$M_{sys2} = M\left(\frac{2r}{L}\right)$

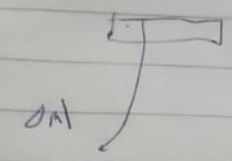
$\frac{dM_{sys2}}{dt} = \frac{M}{L}\,\frac{gt}{2}$

$\frac{gt^2}{4}$

camlin
Date :

Momentum Analysis For System 2

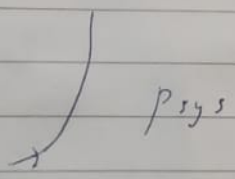
Time t :



$P_{sys} = \cdots + \Delta m(-gt)$

Mass dm falling with this velocity

Time $t + \Delta t$



$P_{sys} = \cdots (gt) \; 0$

Mass Entering with initial velocity

$$F_{ext} = \frac{\Delta m}{\Delta t}(+gt) = \frac{M}{L}\frac{g^2t^2}{2}.$$

$$F_{Table} \; \frac{Mr}{L} g = \frac{M}{L}\frac{g^2t^2}{2}$$

$$F_{Table} = \frac{Mg^2t^2}{4L} + \frac{Mg^2t^2}{2L}$$

$$= \frac{3Mg^2t^2}{4L}$$